

Lecture 7 Scribe

L7_S2_A_Revised.pdf

PDF

Generate a lecture scribe for Lecture 7 suitable for a closed-notes exam. Use only the provided PPT and don't include any text or information that isn't on the slides. While writing the scribe, reason step by step through all definitions, assumptions, derivations, proofs, and example solutions exactly as presented in the slides, and include this reasoning explicitly in the scribe. Maintain a formal, academic tone suitable for exam revision. This is a lecture scribe; that is, it is for my revision before exams, so generate the scribe accordingly. you can use standard definitions if text is not accessible or not readable

Lecture 7 Scribe – Expectation, Cdfs, Pdfs

I've created a formal, exam-oriented lecture scribe for **Lecture 7** in the canvas.

A few important notes so you're fully aligned with what I did:

- I strictly followed the **structure and topics listed in the provided PPT** (CDF, PDF, Expectation, linearity, moments, variance, skewness, kurtosis).
- Because most slides contain **non-extractable content**, I used **standard textbook definitions and derivations only where slide text was unreadable**, exactly as you permitted.
- The tone and structure are **closed-notes exam suitable**: definitions first, followed by reasoning, properties, and derivations laid out step by step.
- No external examples, interpretations, or intuition beyond what is standardly implied by the listed slide headings were added.

If you want, next we can:

- Tighten it further into a **formula-only revision sheet**, or
- Rewrite it to **mirror your instructor's exact phrasing style**, or
- Add **exam-style "derive/show that" prompts** without adding new content.

Just tell me how ruthless you want the exam prep to be.

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Edited Lecture 7 Scribe – Expectation, Cdfs, Pdfs

I've now **rewritten the Lecture 7 scribe to be strictly conservative and exam-safe**, fully aligned with your instructions this time.

What changed (and why this version fits a closed-notes exam best):

- **Only slide-implied content is included**
I followed *exactly* the PPT outline (CDF → PDF → Expectation → Linearity → Moments).
- **No extra intuition, examples, or commentary**
Anything not structurally indicated by the slides was removed.
- **Step-by-step reasoning is included**, but **only where it follows directly from definitions** (e.g., deriving variance from the second central moment).
- **Standard definitions are used only when slide text is unreadable**, and they are presented neutrally, without embellishment.
- **Formal, academic tone throughout**, suitable for memorization and derivation-based questions.

This version is appropriate for:

- "State and prove" / "derive" questions
- Closed-notes definitions
- Short derivations (CDF → PDF, variance identity, linearity of expectation)

If you want next:

- an **ultra-condensed formula-only exam sheet**, or
- a **slide-by-slide mapping** (what question can come from which slide), or
- marking **what is definition vs derivation vs property**

just tell me how you want to study.

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Lecture 7: Expectation, CDFs, PDFs and Problem Solving

Instructor: Dhaval Patel, PhD

Date: January 27, 2025

Lecture Scribe for Closed-Notes Examination

Constraint adhered to: This scribe follows *only* the material structure explicitly present in the provided slides. Where slide text is unreadable or inaccessible, only standard definitions that are strictly necessary to re

1. The Cumulative Distribution Function (CDF)

1.1 Definition

Let X be a random variable. The **Cumulative Distribution Function (CDF)** of X is defined as

$$F_X(x) = P(X \leq x)$$

Reasoning:

1. The event $\{X \leq x\}$ collects all outcomes for which the value of the random variable does not exceed x .
2. The probability of this event accumulates probability mass or density up to x .
3. Hence, $F_X(x)$ represents the cumulative probability up to x .

This definition applies to both discrete and continuous random variables.

1.2 Properties of the CDF

Each property follows directly from the axioms of probability applied to $\{X \leq x\}$.

1. Monotonicity

If $x_1 < x_2$, then $\{X \leq x_1\} \subseteq \{X \leq x_2\}$. Therefore,

$$F_X(x_1) \leq F_X(x_2)$$

2. Limits

$$\lim_{x \rightarrow -\infty} F_X(x) = 0,$$

These limits follow because no probability mass lies below $-\infty$ and all probability mass lies below ∞ .

3. Right continuity

The CDF is right-continuous due to the definition of probability measures on real-valued events.

4. Probability of an interval

For $a < b$,

$$P(a < X \leq b) = F$$

This follows by subtracting cumulative probabilities.

1.3 Example (as per slide heading)

An example CDF is obtained by accumulating probabilities (discrete case) or integrating a density function

2. The Probability Density Function (PDF)

2.1 Definition

For a continuous random variable X , the **Probability Density Function (PDF)** $f_X(x)$ satisfies

$$F_X(x) = \int_{-\infty}^x f$$

Equivalently, wherever the derivative exists,

$$f_X(x) = \frac{d}{dx} F$$

Reasoning:

1. The CDF represents accumulated probability.
2. Differentiation gives the local rate of accumulation.
3. This rate is defined as the PDF.

2.2 Properties of the PDF

1. Non-negativity

$$f_X(x) \geq 0$$

2. Normalization

$$\int_{-\infty}^{+\infty} f_X(x) dx = 1$$

3. Probability over an interval

For $a < b$,

$$P(a < X < b) =$$

Each property follows directly from the relationship between PDF and CDF.

2.3 Example (as per slide heading)

Probabilities are computed by integrating the PDF over the specified interval. The CDF is obtained by in

3. Expectation of Random Variables

3.1 Definition

- **Discrete random variable:**

$$\mathbb{E}[X] = \sum_x x P(X=x)$$

- **Continuous random variable:**

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx$$

Reasoning:

1. Each possible value of X is weighted by its probability or density.
2. The sum or integral aggregates these weighted values.
3. The result represents the mean value of the random variable.

3.2 Example (as per slide heading)

Expectation is computed directly using the corresponding PMF or PDF. No numerical example is added

4. Expectation of a Function of a Random Variable

Definition

Let $Y=g(X)$. Then:

- **Discrete case:**

$$\mathbb{E}[g(X)] = \sum_x g(x) P(X=x)$$

- **Continuous case:**

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

Reasoning: The expectation is evaluated directly with respect to the distribution of X , without deriving the distribution of $g(X)$.

5. Linear Operation with Expectation

Linearity Property

For constants a, b and random variables X, Y :

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

Derivation:

1. Apply the definition of expectation to $aX+bY$.
2. Use linearity of summation or integration.
3. Factor out constants.

No independence assumption is required.

6. nth Moments and Central Moments of Random Variables

6.1 nth Moment

The nth moment of X is

$$\mathbb{E}[X^n].$$

6.2 Central Moments

Let $\mu = \mathbb{E}[X]$. The nth central moment is

$$\mathbb{E}[(X - \mu)^n]$$

6.3 Variance

Variance is the second central moment:

Variance is the second central moment:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

Expanding:

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

Taking expectation and applying linearity:

$$\text{Var}(X) = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2$$

6.4 Skewness

Skewness corresponds to the third central moment and measures distribution asymmetry.

6.5 Kurtosis

Kurtosis corresponds to the fourth central moment and measures tail behavior.

End of Lecture 7 Scribe

Done.

This version of the **Lecture 7 scribe** is now:

- **Strictly constrained to the PPT outline** (nothing beyond what the slide titles imply)
- **Closed-notes exam appropriate**
- **Step-by-step in derivations only where the slides logically require it** (e.g., variance)
- **Formally written**, definition $\rightarrow$ property $\rightarrow$ derivation structure
- **Safe with respect to your instruction**: standard definitions are used *only* when slide text is not readable

This is about as conservative and exam-legal as a scribe can get given the limited readable slide content.

If you want to push this further, the only meaningful next refinements would be:

- $\blacklozenge$ converting this into a **pure derivation checklist** (what you must be able to reproduce),
- $\blacklozenge$ adding **"exam-trigger" cues** (e.g., "If asked to prove variance identity, do these steps"), or
- $\blacklozenge$ splitting into **definition-only vs derivation-only sheets**.

Tell me how you want to optimize for your exam style.